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(57) Abstract :

Renewable energy has distributed generation as significant challenge where energy generation and consumption are generally located in nearness. A solution is provided by smartgrid for this challenge as it avails integration of distributed generation reliably which includes storage of energy and controlled loads. Smartgrid is a electricity generation system that operates in bidirectional mode allowing electricity distribution from suppliers to consumers through digital technology. Hence Smartgrids boosts integration of renewable energy sources thereby optimizing energy management through machine learning algorithm. This invention designs and develops wireless sensor network system in integration with webbased platform for optimal management of smartgrid through continuous monitoring. High capability of data processing with high storage capacity is availed by integrating machine learning algorithm and IoT platform with LoRaWan technology for deploying and implementing low power wireless remote monitoring network for optimized management of smartgrids.

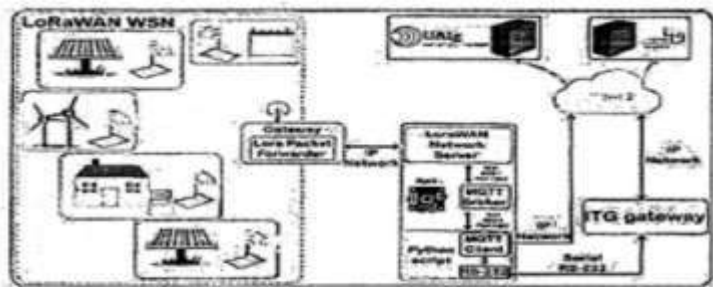


Figure 1. Proposed Architecture of Machine Learning-based Energy Management of Smartgrids